

Rahul Chandran Lakshmanan

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EDUCATION

Bachelors of Science, Computer Science

September 2020 - June 2024

University of California, Davis | Dean's Honors List | GPA: 3.65

WORK EXPERIENCE

Software Developer

May 2025 - Current

Vast Vertex Media

- Developing a GenAI podcast application enabling users to generate customized podcast audio with a focus on Major League Baseball (MLB).
- Building a Model Context Protocol (MCP) server in Python that enables the application's LLM to automatically call external MLB APIs and retrieve relevant sports data in real time based on user provided context.
- Engineered reusable, context aware prompt templates using advanced prompt engineering techniques.
- Utilized Python scripting to fetch MLB data from external sources via REST API calls and integrate them into prompt templates to support accurate and real time content generation.
- Developed a lightweight React frontend that enables users to define custom templates and generate podcast scripts via an intuitive and user-friendly interface.
- Automated backend workflows in Java with Hibernate and PostgreSQL to efficiently manage and store generated content.

Associate Software Engineer Intern

September 2024 - April 2025

Altimetrik

- Developed a RAG-based Jira chatbot to enable natural language queries over custom Jira data, leveraging data retrieval and LLM generated responses to streamline data access, enhance project analysis, and refine decision making.
- Integrated Amazon Bedrock, OpenSearch Serverless, and S3 with LangChain Python framework to construct vector stores and implement query based data filtering, enabling efficient data context retrieval for accurate LLM response generation.
- Utilized DynamoDB to store user chat interactions and application data, supporting features like chat history persistence and past context aware interactions.
- Built a chat interface UI with Streamlit Python framework, allowing conversational interactions with the LLM and enhancing the querying experience through context aware follow up capabilities.
- Deployed a CI/CD pipeline with Argo Workflows, Amazon EKS, and ECR to automate Docker image builds upon repository updates, facilitating seamless integration with the deployment environment.
- Leveraged Jira for task and project tracking within a Scrum framework, facilitating agile development.
- Utilized Bitbucket and Git for version control, code hosting, and collaboration, ensuring streamlined workflows and efficient team coordination.

PROJECT EXPERIENCE

Enigma Machine

- Developed software simulation of Enigma machine in C++, enabling accurate encryption and decryption of plaintext.
- Utilized GoogleTest framework, generated shared library, and employed CMake and Docker for the build system.

Spotifind

- Developed an interactive music discovery app using SwiftUI and Spotify Web API.
- Coded a web version using HTML/CSS/Javascript for frontend and Python/Flask for backend.
- Implemented dynamic song recommendations utilizing user preferences and tinder-like swiping feedback mechanisms.

SKILLS

Languages: Python, C/C++, Java, C#, HTML, CSS, JavaScript, React, SQL

Technologies: Git/GitHub/Bitbucket, Docker, Amazon Web Services, Argo Workflows, Kubernetes, Jira